Ashwin Deshpande

Data Science Researcher with experience applying econometrics, statistical modeling, and time-series analysis to macroeconomic and financial datasets. Currently conducting policy research at the Reserve Bank of India on commodity price transmission, inflation dynamics, and market integration using autoregressive methods to understand pricing dynamics. Experienced developing reproducible analytical workflows, forecasting economic indicators, and communicating quantitative findings through technical reports and stakeholder presentations. Proficient in Python, R, SQL, and modern statistical methods for economic research.

CONTACT

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Github: https://github.com/AshwinDeshpande96 **Technical blog:** https://ashwindeshpande96.github.io/personal_webpage/

PROFESSIONAL EXPERIENCE

Reserve Bank of India - Data Science Lab, DSIM | **Pre-doc Researcher** | *Mumbai, India* — *Jan 2026 – June 2026*

- Conducted quantitative research on spatio-temporal and vertical price transmission across agricultural supply chains, examining inflation dynamics, market integration, and commodity price behavior.
- Applied econometric and time-series methods including Bayesian VAR, Unrestricted Error Correction Models (UECM), SARIMA-GARCH, and structural break analysis to evaluate shock persistence, seasonality, and forecasting performance.
- Built analytical datasets, performed statistical analysis, and prepared technical reports and visualizations to support policy discussions on inflation, supply chains, and agricultural markets. [Working Paper] [Slides]

Thynk Capital LLC | **Data Scientist** | *Remote, USA* — *May 2023 – Feb 2025*

- Developed predictive models and quantitative research pipelines using large financial time-series datasets to support systematic investment decisions across multiple asset classes.
- Designed reproducible data processing and forecasting workflows using Python, SQL, PySpark, Airflow, and MLflow, enabling scalable model development and evaluation.
- Performed exploratory data analysis, feature engineering, and backtesting to assess model performance under varying market conditions and volatility regimes.

Bombora Inc | **Data Scientist Intern** | *Remote, USA* — *Jun 2022 – Dec 2022*

- Built machine learning pipelines to process millions of company records and support large-scale predictive analytics.
- Undertook statistical analysis, feature engineering, and model evaluation to improve classification accuracy and reduce false positives.
- Developed dashboards and analytical reports to communicate model insights to technical and business stakeholders.

UIC NLP Lab | **Graduate Research Assistant** | *Chicago, IL* — *Jun 2022 – Dec 2022*

- Performed NLP research using statistical and deep learning methods to study speech biomarkers for dementia detection.
- Developed reproducible research pipelines and co-authored a peer-reviewed publication at ACL BioNLP 2022: [Paper]

Primenumbers Technologies | **Data Science Engineer** | *Bengaluru, India* — *May 2019 - May 2020*

- Built ETL pipelines and analytical workflows using Python and PostgreSQL to process large government procurement datasets.
- Developed machine learning models and evaluated their performance through statistical analysis and A/B testing.

PROJECTS

- Survival Analysis: Time-to-event analysis using hierarchical mixture models (*Bayesian Inference | Survival Model*) [Blog]
- Product recommendation for potential banking customers for increasing revenue (*Expected Revenue | Propensity Model*) [Code][Slides]

SKILLS

Programming: Python, R, SQL, PySpark, Git, MATLAB, STATA, Tableau, PowerBI

Econometrics: Quantile Nonlinear ARDL, Spatial-Temporal Shock Transmission, BVAR, VECM, Statistical Inference.

Research: Economic Data Analysis, Policy Research, Data Visualization, Technical Writing, Literature Review

EDUCATION

University of Illinois, Chicago — *M.S. in Computer Science - 4.0 GPA* - 2022
B.V.B College of Engineering & Technology, Hubli — *B.Eng in Computer Science - 3.2 GPA* - 2018

CERTIFICATES

Deeplearning.ai-Agentic AI Deep Learning.ai-Intro to MLOps Deeplearning.ai-ML Data Lifecycle Linear Algebra Bayesian Statistics - L1